PROJECT: Hybrid Cloud Setup Using VPN or Direct Connect

Services: AWS Site-to-Site VPN, VPC, EC2, Route Tables
Goal: Connect an on-premises simulated environment (e.g., using VirtualBox) with AWS.
Skills: Routing, hybrid DNS, secure connectivity.

Definition: Hybrid Cloud Setup Using VPN or Direct Connect refers to a network architecture where an organization connects its on-premises infrastructure (like a private data center or local network) to a public cloud environment (like AWS, Azure, or GCP) in a secure and seamless way, allowing workloads to communicate across both environments.
Scope of the Project: The scope of this project focuses on designing and implementing a secure hybrid cloud environment by connecting a simulated on-premises network with AWS infrastructure. The project includes creating an AWS VPC, launching EC2 instances, configuring routing and security rules, and establishing a Site-to-Site VPN connection between the on-premises router and AWS Virtual Private Gateway. The scope also covers testing secure connectivity between the two networks and demonstrating how hybrid cloud architectures can be used to extend enterprise networks into the cloud for improved scalability, flexibility, and security.

Methodologies:

1. **Method used in this Project:** The scope of this project focuses on designing and implementing a secure hybrid cloud environment by connecting a simulated on-premises network with AWS infrastructure.
   • The project includes creating an AWS VPC, launching EC2 instances, configuring routing and security rules, and establishing a Site-to-Site VPN connection between the on-premises router and AWS Virtual Private Gateway.
   • The scope also covers testing secure connectivity between the two networks and demonstrating how hybrid cloud architectures can be used to extend enterprise networks into the cloud for improved scalability, flexibility, and security.

Services Used in this Project:

1. **Amazon VPC (Virtual Private Cloud)**
   • **Definition:** A Virtual Private Cloud is a logically isolated network in AWS where cloud resources like EC2 instances can run securely.
   • **Purpose:** The VPC is used to create a private cloud network to host the EC2 server that connects to the on-premises network through the VPN.
   • **Role in this Project:** The VPC acts as the AWS-side network that communicates with the on premises network through the Site-to-Site VPN tunnel.

2. **Amazon EC2 (Elastic Compute Cloud)**

   • **Definition:** Amazon EC2 is a service that provides virtual servers in AWS to run applications and workloads.

   • **Purpose:** EC2 is used to create a cloud server inside the AWS VPC to test connectivity with the on premises network through the VPN.

   • **Role in this Project:** EC2 acts as the cloud-side machine that communicates with the on-premises environment through the Site-to-Site VPN tunnel.

3. **Virtual Private Gateway**

   • **Definition:** A Virtual Private Gateway is an AWS gateway that enables secure VPN connectivity between an AWS VPC and an on-premises network.

   • **Purpose:** The VGW is used to connect the AWS VPC to the on-premises network through the Site to-Site VPN tunnel.

   • **Role in this Project:** The VGW acts as the AWS-side endpoint that receives and routes VPN traffic between the VPC and the on-premises network.

4. **Customer Gateway**

   • **Definition:** A Customer Gateway is a resource in AWS that represents the on-premises VPN device used to connect to AWS VPC.

   • **Purpose:** The CGW is used to define the on-premises router's public IP address to establish the Site to-Site VPN connection with AWS.

   • **Role in this Project:** The CGW acts as the on-premises endpoint that creates the secure VPN tunnel with the AWS Virtual Private Gateway.

5. **AWS site-to-site VPN**

   • **Definition:** AWS Site-to-Site VPN is a service that creates a secure IPsec encrypted tunnel between an on-premises network and an AWS VPC over the internet.

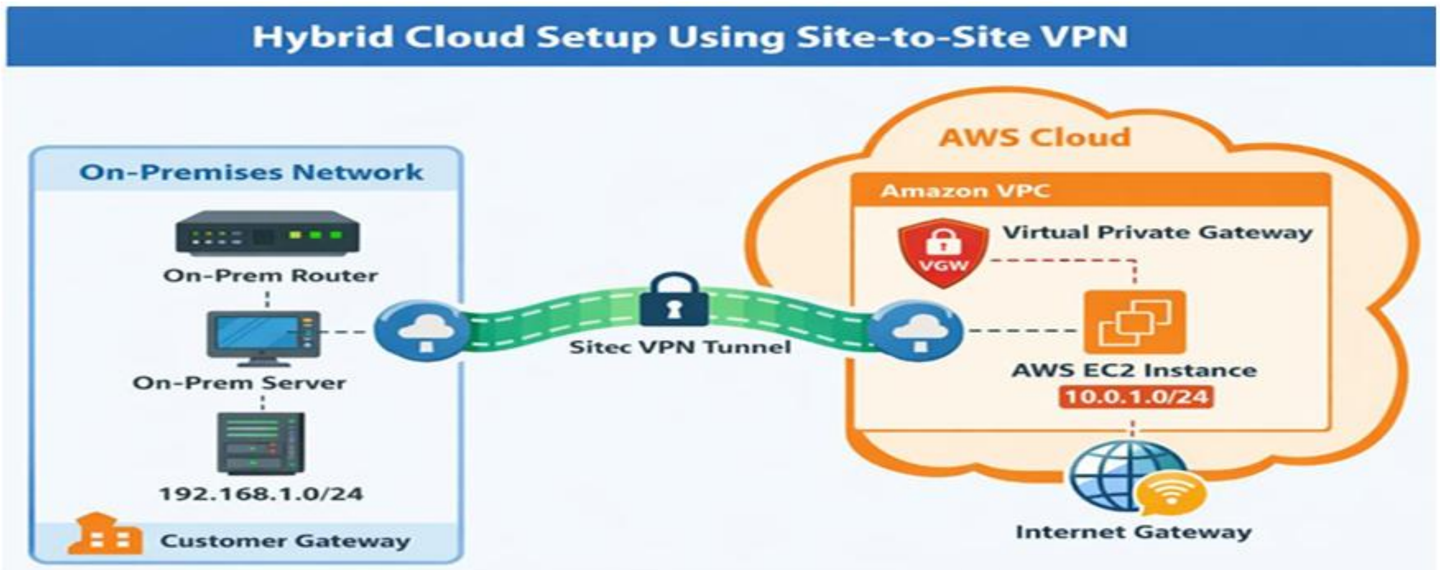
   • **Purpose:** It is used to securely connect the on-premises simulated network with the AWS VPC network.

   • **Role in this Project:** It acts as the secure communication tunnel that allows data to travel safely between the on-prem router and the AWS cloud server.

6. Route Table

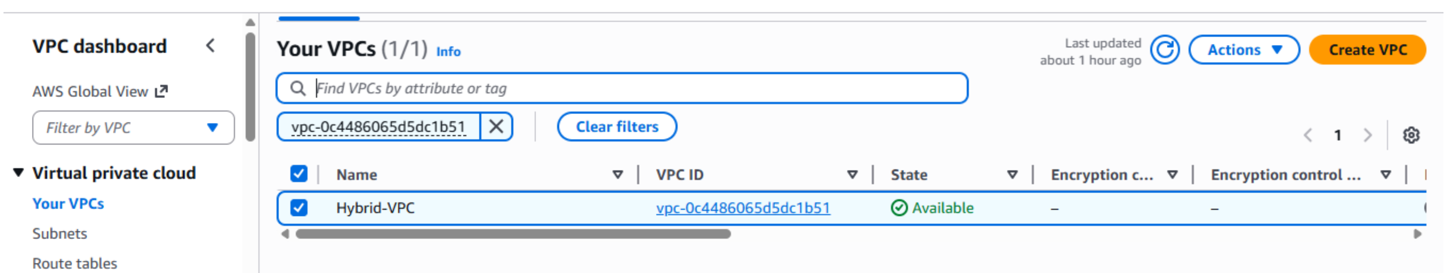
- **Definition:** A Route Table is a set of rules in AWS that determines how network traffic is directed within a VPC.
- **Purpose:** The route table is used to direct traffic between the AWS VPC network and the on premises network through the VPN gateway.
- **Role in this Project:** The route table ensures that traffic destined for the on-prem network (192.168.1.0/24) is routed through the Virtual Private Gateway over the VPN tunnel.

Architecture Diagram:

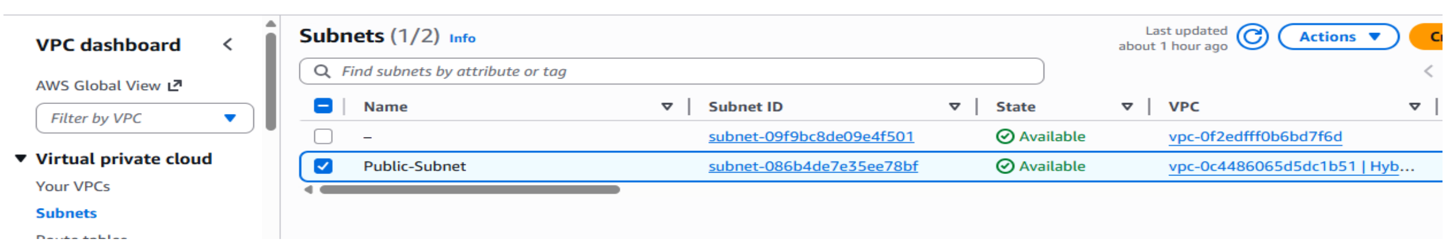


Implementation and Execution of Hybrid Cloud Setup Using Site-to-Site VPN.

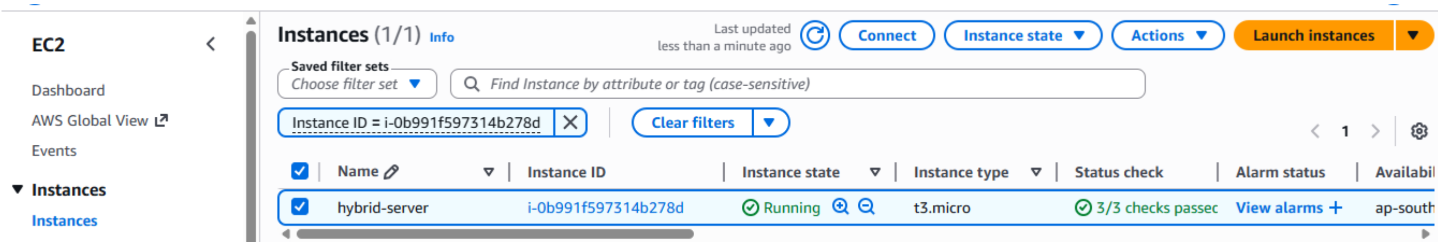
STEP 1: Created VPC.



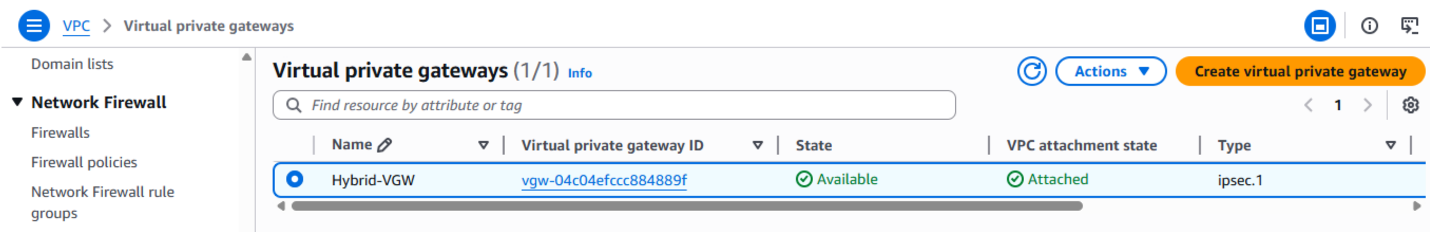
STEP 2: Created Subnet.



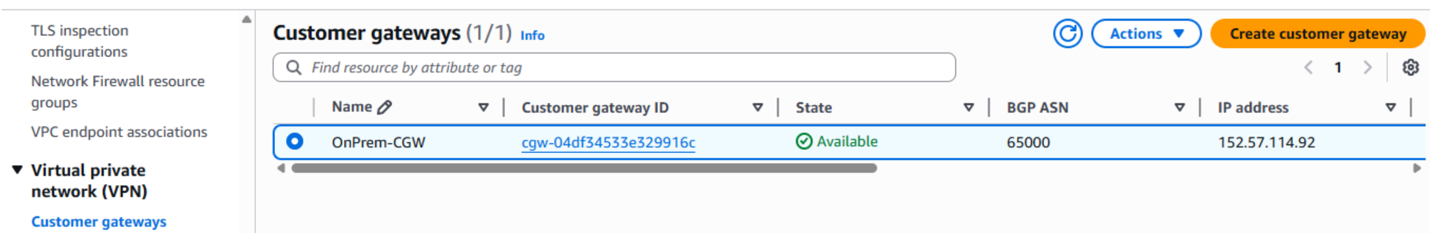
STEP 3: Launched and instance.



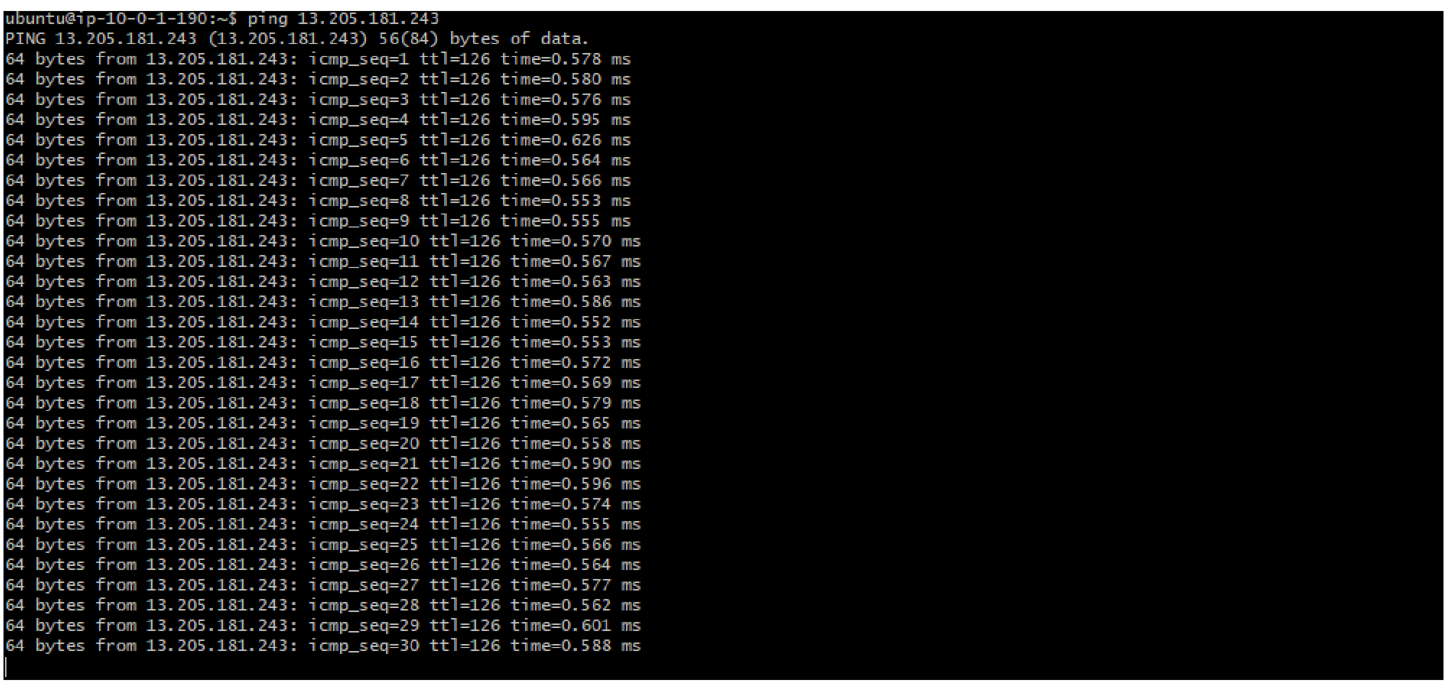
STEP 4: Created Virtual Private Gateway.



STEP 5: Created Customer Gateway.



STEP 6: Final Output.



Troubleshoot:

- In virtual box installation of ubuntu is took lot of time
- so, I connect the server in gitbash

Conclusion:

This project successfully demonstrated a Hybrid Cloud setup by connecting an on-premises network with an AWS VPC using a Site-to-Site VPN. A secure IPsec tunnel was established between the on-prem router and the AWS Virtual Private Gateway to enable private communication between both networks. The connectivity was verified by successfully transmitting data between the on-prem environment and the AWS EC2 instance. This setup shows how organizations can securely extend their local infrastructure to the cloud. Overall, the project highlights the effectiveness of VPN technology in building scalable and secure hybrid cloud environments.